Linux Text Processing Task: grep, cut, awk

How to do this task

Your team answers all 30 questions using only `grep`, `cut` and `awk`, together with `sort`, `uniq`, `wc` and pipes where they help.

1. Make your own copy of this doc and name it `<Team name> – grep cut awk task`.
2. On your EC2 instance, create the log file: run `vi access.log`, paste the 50 lines from the Sample log section, then save.
3. Check that the file has 50 lines: `wc -l access.log`.
4. For every question, write the **command** you ran and paste the **output** in the space below it.
5. Rotate the Driver after every 5 questions so everyone types commands.
6. The Scribe shares the finished doc in reply to the team email by 10:00 PM.

All questions use the file `access.log`.

[DELETE ABOVE AFTER YOU COPIED]

Team Name:
Attendance: no of members attended

Sample log (access.log)

The log is a web server access log. Each line holds: client IP, timestamp, request (method, path, protocol), status code, response size in bytes, and user agent.

```
203.0.113.45 - - [05/Nov/2026:10:01:01 +0530] "GET /missing.html HTTP/1.1" 404 243
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:01:48 +0530] "GET /api/orders HTTP/1.1" 500 330
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:02:36 +0530] "GET /about.html HTTP/1.1" 200 10255
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:03:49 +0530] "POST /api/login HTTP/1.1" 200 10153
"Googlebot/2.1"
192.168.1.15 - - [05/Nov/2026:10:04:33 +0530] "GET /images/logo.png HTTP/1.1" 200
20573 "curl/8.4.0"
192.168.1.15 - - [05/Nov/2026:10:05:51 +0530] "PUT /api/users/12 HTTP/1.1" 200 13815
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:07:08 +0530] "GET /images/logo.png HTTP/1.1" 200
21454 "Mozilla/5.0"
10.0.0.8 - - [05/Nov/2026:10:07:37 +0530] "GET /Admin/settings HTTP/1.1" 403 327
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:08:31 +0530] "GET /images/logo.png HTTP/1.1" 200
21160 "Googlebot/2.1"
```

```
192.168.1.10 - - [05/Nov/2026:10:09:35 +0530] "POST /api/orders HTTP/1.1" 503 447
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:10:26 +0530] "GET /contact.html HTTP/1.1" 200 2120
"python-requests/2.31"
10.0.0.5 - - [05/Nov/2026:10:11:21 +0530] "GET /old-page HTTP/1.1" 301 209
"Googlebot/2.1"
192.168.1.10 - - [05/Nov/2026:10:12:10 +0530] "GET /api/users HTTP/1.1" 200 5104
"python-requests/2.31"
203.0.113.45 - - [05/Nov/2026:10:13:10 +0530] "GET /favicon.ico HTTP/1.1" 404 293
"curl/8.4.0"
192.168.1.10 - - [05/Nov/2026:10:14:20 +0530] "GET /missing.html HTTP/1.1" 404 191
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:14:46 +0530] "POST /api/login HTTP/1.1" 401 257
"curl/8.4.0"
10.0.0.5 - - [05/Nov/2026:10:15:25 +0530] "GET /old-page HTTP/1.1" 301 271
"curl/8.4.0"
192.168.1.10 - - [05/Nov/2026:10:16:24 +0530] "POST /api/login HTTP/1.1" 200 13082
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:17:51 +0530] "GET /index.html HTTP/1.1" 200 13221
"Mozilla/5.0"
10.0.0.8 - - [05/Nov/2026:10:18:56 +0530] "GET /favicon.ico HTTP/1.1" 404 199
"Googlebot/2.1"
192.168.1.15 - - [05/Nov/2026:10:19:45 +0530] "GET /contact.html HTTP/1.1" 200 6625
"Googlebot/2.1"
192.168.1.22 - - [05/Nov/2026:10:20:49 +0530] "GET /old-page HTTP/1.1" 301 206
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:22:10 +0530] "GET /index.html HTTP/1.1" 200 8655
"Googlebot/2.1"
192.168.1.22 - - [05/Nov/2026:10:22:55 +0530] "GET /api/users HTTP/1.1" 200 7909
"Googlebot/2.1"
10.0.0.8 - - [05/Nov/2026:10:23:25 +0530] "GET /api/users HTTP/1.1" 200 2881
"curl/8.4.0"
192.168.1.10 - - [05/Nov/2026:10:24:29 +0530] "POST /api/login HTTP/1.1" 401 234
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:24:52 +0530] "GET /admin HTTP/1.1" 403 266
"Mozilla/5.0"
172.16.0.3 - - [05/Nov/2026:10:25:57 +0530] "GET /favicon.ico HTTP/1.1" 404 284
"curl/8.4.0"
10.0.0.5 - - [05/Nov/2026:10:26:40 +0530] "POST /api/login HTTP/1.1" 401 258
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:28:06 +0530] "GET /contact.html HTTP/1.1" 200 13649
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:28:38 +0530] "GET /index.html HTTP/1.1" 200 13251
"Mozilla/5.0"
192.168.1.22 - - [05/Nov/2026:10:29:33 +0530] "GET /contact.html HTTP/1.1" 200 9119
"curl/8.4.0"
192.168.1.15 - - [05/Nov/2026:10:30:08 +0530] "GET /Admin/settings HTTP/1.1" 403 218
"Mozilla/5.0"
203.0.113.45 - - [05/Nov/2026:10:30:47 +0530] "GET /old-page HTTP/1.1" 301 185
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:31:35 +0530] "GET /api/orders HTTP/1.1" 500 463
"Googlebot/2.1"
192.168.1.10 - - [05/Nov/2026:10:32:41 +0530] "POST /api/orders HTTP/1.1" 503 451
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:33:09 +0530] "PUT /api/users/12 HTTP/1.1" 200 2516
```

```
"Mozilla/5.0"
172.16.0.3 - - [05/Nov/2026:10:34:23 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:35:18 +0530] "GET /api/users HTTP/1.1" 200 7768
"python-requests/2.31"
192.168.1.10 - - [05/Nov/2026:10:36:06 +0530] "POST /api/login HTTP/1.1" 200 8234
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:37:33 +0530] "POST /api/login HTTP/1.1" 200 3445
"python-requests/2.31"
192.168.1.15 - - [05/Nov/2026:10:38:19 +0530] "GET /favicon.ico HTTP/1.1" 404 214
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:39:43 +0530] "POST /api/users HTTP/1.1" 201 443
"python-requests/2.31"
203.0.113.45 - - [05/Nov/2026:10:40:53 +0530] "GET /admin HTTP/1.1" 403 250
"Mozilla/5.0"
192.168.1.10 - - [05/Nov/2026:10:41:29 +0530] "DELETE /api/users/7 HTTP/1.1" 204 0
"Mozilla/5.0"
10.0.0.5 - - [05/Nov/2026:10:42:20 +0530] "GET /about.html HTTP/1.1" 200 8327
"Mozilla/5.0"
192.168.1.15 - - [05/Nov/2026:10:43:50 +0530] "GET /about.html HTTP/1.1" 200 5871
"curl/8.4.0"
192.168.1.22 - - [05/Nov/2026:10:44:45 +0530] "POST /api/users HTTP/1.1" 201 458
"python-requests/2.31"
192.168.1.15 - - [05/Nov/2026:10:45:10 +0530] "GET /admin HTTP/1.1" 403 349
"Googlebot/2.1"
10.0.0.8 - - [05/Nov/2026:10:46:19 +0530] "GET /contact.html HTTP/1.1" 200 3248
"Googlebot/2.1"
```

Section A: grep (Questions 1 to 10)

Q1. Show every request that returned status code 404.

Command:

Output:

Q2. Count how many POST requests are in the log.

Command:

Output:

Q3. Show all requests made from the IP address 192.168.1.15.

Command:

Output:

Q4. Show every line that does NOT have status code 200.

Command:

Output:

Q5. Find every request to an admin page, whether it is written as "admin" or "Admin" (case-insensitive search).

Command:

Output:

Q6. Show the line numbers of all requests that returned status 500.

Command:

Output:

Q7. Show all requests made using the curl user agent.

Command:

Output:

Q8. Show all requests to API endpoints (paths starting with /api/).

Command:

Output:

Q9. Count all error responses, meaning any 4xx or 5xx status code. Hint: use an extended regular expression.

Command:

Output:

Q10. Print only the IP addresses that start with 10. (print just the matching part, not the whole line).

Command:

Output:

Section B: cut (Questions 11 to 20)

Q11. Print only the IP address from every line.

Command:

Output:

Q12. Print the IP address and the status code of every request.

Command:

Output:

Q13. Print only the request part that sits inside the first pair of double quotes, for example `GET /index.html HTTP/1.1`.

Command:

Output:

Q14. Print the full timestamp without the square brackets, for example `05/Nov/2026:10:01:01 +0530`.

Command:

Output:

Q15. Print only the time of each request in HH:MM:SS format, for example `10:01:01`.

Command:

Output:

Q16. Print only the HTTP method (GET, POST, PUT or DELETE) of each request.

Command:

Output:

Q17. Print the first 15 characters of every line.

Command:

Output:

Q18. Print the list of unique IP addresses, sorted.

Command:

Output:

Q19. Count how many requests came from each IP address, with the busiest IP at the top.

Command:

Output:

Q20. Print only the user agent of each request, for example `Mozilla/5.0`.

Command:

Output:

Section C: awk (Questions 21 to 30)

Q21. Print the IP address and the requested path of every line.

Command:

Output:

Q22. Print only the lines where the status code is exactly 500.

Command:

Output:

Q23. Calculate the total number of bytes sent across all requests.

Command:

Output:

Q24. Count how many requests there are for each status code.

Command:

Output:

Q25. Print the IP, path and size of every response larger than 10000 bytes.

Command:

Output:

Q26. Calculate the average response size in bytes.

Command:

Output:

Q27. Print each line with its line number in front, then print the total number of lines at the end.

Command:

Output:

Q28. For POST requests only, print the path and the status code.

Command:

Output:

Q29. Find the single IP address that made the most requests, and print it with its count.

Command:

Output:

Q30. Use printf to print a neat report line for every request, in this format: `IP: 10.0.0.5 | Status: 200 | Size: 10.01 KB` (size in KB with 2 decimals).

Command:

Output:

Bonus (optional)

Using one awk command, print the number of failed login attempts (POST /api/login with status 401) for each IP address.